

Experiment No. 1

Experiment Title: Air-Quality Data Cleaning Using R: Management of NA, NULL, and NaN Values

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Problem Statement:

An environmental monitoring agency collects hourly air-quality and weather readings from multiple monitoring stations. Due to sensor failures, temporary network interruptions, equipment maintenance, and data-entry errors, several pollutant and weather readings are unavailable or invalid. Before this data can be used for pollution analysis, it must be cleaned and validated.

This practical uses the **Beijing Multi-Site Air Quality Dataset** (specifically the Aotizhongxin station, PRSA_Data_Aotizhongxin_20130301-20170228.csv), which contains 35,064 hourly observations across 18 variables including PM2.5, PM10, SO2, NO2, CO, O3, TEMP, PRES, DEWP, RAIN, wd, and WSPM. Some observations contain missing values represented as NA.

The objective is to develop an R program that can:

- Identify missing and undefined values (NA, NULL, NaN)
- Differentiate between these three types of missing/undefined data
- Summarize missing values across selected variables
- Replace missing numerical values using suitable statistical methods
- Replace missing categorical values appropriately
- Handle errors gracefully without terminating the program
- Export a cleaned dataset for further analysis

Task:

1. Import and Inspect the Dataset
2. Understand NA, NULL, and NaN
3. Create a Missing-Value Summary Function
4. Identify Invalid Numerical Results
5. Handle Missing Numerical Values Using a Loop
6. Handle Missing Categorical Values
7. Implement Error Handling
8. Compare Missing Values Before and After Cleaning
9. Generate One Visualization
10. Export the Cleaned Dataset

GitHub Link: <https://github.com/rishirp15/R-Programming/tree/main/Experiment%201>

Missing-Value Summary Table (Task 3):

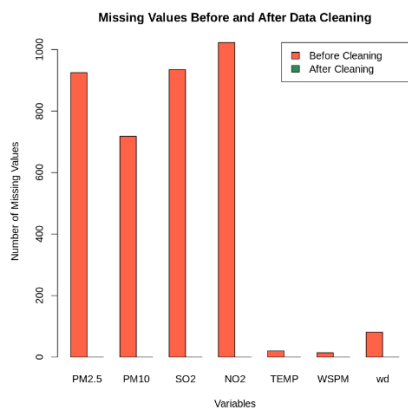
	Variable	Total_Records	Missing_Values	Missing_Percentage
1	PM2.5	35064	925	2.64
2	PM10	35064	718	2.05
3	SO2	35064	935	2.67
4	NO2	35064	1023	2.92
5	TEMP	35064	20	0.06
6	WSPM	35064	14	0.04
7	wd	35064	81	0.23

Comparison Table — Before vs After Cleaning (Task 8)

	Variable	Missing_Before	Missing_After	Values_Replaced
PM2.5	PM2.5	925	0	925
PM10	PM10	718	0	718
SO2	SO2	935	0	935
NO2	NO2	1023	0	1023
TEMP	TEMP	20	0	20
WSPM	WSPM	14	0	14
	wd	81	0	81

Interpretation: All selected variables show zero missing values after cleaning, confirming that the imputation (median for numeric, mode for categorical) was successful.

Visualization:



Cleaned Dataset: Exported successfully as `cleaned_air_quality_data.csv` using `write.csv()` with `row.names = FALSE`. (uploaded on github)

Interpretation:

The Beijing air-quality dataset (Aotizhongxin station) initially contained 7,271 missing values across 35,064 hourly records. Among the selected variables, missingness ranged from as low as 0.04% (WSPM) to 2.92% (NO2), with none exceeding the 20% warning threshold. Numerical variables — PM2.5, PM10, SO2, NO2, TEMP, and WSPM — were cleaned using a loop-based function that replaced missing entries with each variable's median (e.g., PM2.5: 58, TEMP: 14.5), a robust choice given the skewed nature of pollution data. The categorical variable wd (wind direction) was imputed using its mode, "NE". A derived variable, `pollution_ratio` (PM2.5/PM10), showed 934 NA values but no NaN or infinite results. The error-handling function correctly caught invalid inputs, such as passing a categorical column or a nonexistent variable, without halting execution. After cleaning, all seven variables showed zero missing values, confirming the pipeline worked as intended.